Class 6 Science – Chapter 10: Living Creatures: Exploring Their Characteristics

Simple English Question–Answer Notes | Based on the uploaded textbook, pages 134–146

इस अध्याय के सभी उपलब्ध पृष्ठों को ध्यान में रखते हुए नीचे पुस्तक में दिए गए प्रश्नों, Think and Answer, Quick Check, गतिविधियों तथा प्रत्येक टॉपिक से महत्वपूर्ण और संभावित परीक्षा-प्रश्नों के उत्तर सरल अंग्रेजी में दिए गए हैं।

10-living-creatures-exploring-their-characteristics.pdf

1. Jenny's Living Room – Living and Non-living Things

Q1. What are living things?

Answer: Living things are organisms that perform life processes such as nutrition, respiration, growth, excretion, reproduction, movement, and response to stimuli.

Q2. What are non-living things?

Answer: Non-living things are things that do not perform all the life processes. Examples are tables, chairs, stones, and books.

Q3. Give four examples of living things.

Answer: Human beings, dogs, trees, and birds are living things.

Q4. Give four examples of non-living things found in a living room.

Answer: A sofa, a table, a television, and a chair are non-living things.

Q5. How can we differentiate between living and non-living things?

Answer: Living things perform life processes, such as nutrition, respiration, growth, excretion, and reproduction. Non-living things do not perform all these processes.

Q6. Is a car a living thing because it moves?

Answer: No. A car is a non-living thing. It moves with the help of an engine and fuel, but it does not perform life processes.

2. What Sets Living Things Apart from Non-living Things?

Living things perform different life processes that help them survive. The main characteristics discussed in this chapter are:

1. Cellular organisation
2. Nutrition
3. Respiration
4. Growth
5. Excretion
6. Reproduction
7. Movement
8. Response to stimuli

Q1. What are life processes?

Answer: Life processes are the activities performed by living organisms to survive and maintain life.

Q2. Name the main life processes of living things.

Answer: The main life processes are nutrition, respiration, growth, excretion, reproduction, movement, and response to stimuli.

Q3. Do non-living things grow?

Answer: Non-living things do not show internal biological growth. Some non-living things, such as crystals, may become larger when more material is added to them from outside.

Q4. Can a non-living thing move?

Answer: Yes, a non-living thing can move when an external force or a machine makes it move. However, living things can show their own biological movements.

3. Cellular Organisation

Q1. What is a cell?

Answer: A cell is the basic structural and functional unit of a living organism.

Q2. What is a tissue?

Answer: A tissue is a group of similar cells that work together to perform a specific function.

Q3. What is an organ?

Answer: An organ is a body part made up of different tissues that work together to perform a specific function.

Q4. What is an organ system?

Answer: An organ system is a group of organs that work together to perform a major function in the body.

Q5. What is an organism?

Answer: An organism is an individual living thing, such as a human being, a dog, or a plant.

Q6. Write the structural organisation of the human body in order.

Answer: Cell → Tissue → Organ → Organ system → Organism.

Q7. What are unicellular organisms?

Answer: Unicellular organisms are living things made up of only one cell. Examples are Amoeba, yeast, and many bacteria.

Q8. What are multicellular organisms?

Answer: Multicellular organisms are living things made up of many cells. Examples are human beings, dogs, and most plants.

Q9. What is the difference between unicellular and multicellular organisms?

Unicellular organisms	Multicellular organisms
Made up of one cell.	Made up of many cells.
One cell performs all essential life processes.	Different cells may perform specialised functions.
Examples: Amoeba and yeast.	Examples: Humans and mango trees.

Q10. Why is a microscope required to observe a cell?

Answer: Most cells are too small to be seen clearly with the naked eye. A microscope magnifies them so that their structures can be observed.

4. Nutrition

Q1. What is nutrition?

Answer: Nutrition is the process by which living organisms obtain or make food and use it for energy, growth, and repair.

Q2. Why do living things need food?

Answer: Living things need food for energy, growth, repair of damaged parts, and carrying out life processes.

Q3. How do green plants obtain their food?

Answer: Green plants make their own food by photosynthesis using sunlight, water, and carbon dioxide.

Q4. What is photosynthesis?

Answer: Photosynthesis is the process by which green plants use sunlight to make food from carbon dioxide and water, releasing oxygen.

Q5. Why are animals dependent on plants or other animals for food?

Answer: Animals cannot make their own food through photosynthesis. They obtain food by eating plants or other animals.

Q6. What is the role of sunlight in photosynthesis?

Answer: Sunlight provides the energy needed by green plants to make food.

Q7. What is the role of leaves in photosynthesis?

Answer: Leaves contain chlorophyll, which absorbs sunlight and helps the plant make food.

Q8. What is the difference between nutrition in plants and animals?

Answer: Green plants make their own food through photosynthesis, while animals obtain food by eating plants or other animals.

5. Respiration

Q1. What is respiration?

Answer: Respiration is the process by which energy is released from food inside living cells.

Q2. Why do living organisms need respiration?

Answer: Living organisms need the energy released during respiration to grow, move, repair their bodies, and perform other life processes.

Q3. What happens during aerobic respiration?

Answer: In aerobic respiration, glucose reacts with oxygen to release energy, carbon dioxide, and water.

Q4. Write the word equation for aerobic respiration.

Answer:

Glucose + Oxygen → Carbon dioxide + Water + Energy

Q5. How is respiration different from breathing?

Answer: Breathing is the process of taking in air and giving out air. Respiration is the chemical process inside cells that releases energy from food.

Q6. Do plants respire?

Answer: Yes. Plants respire to release energy from food, just as animals do.

Q7. How do humans and many other animals exchange gases?

Answer: Humans and many other land animals exchange gases through their respiratory systems, including the lungs.

Q8. How do plants exchange gases?

Answer: Plants exchange gases through small openings called stomata, which are found mainly on the underside of leaves. Gases can also move through other plant surfaces.

Q9. How do fish obtain oxygen?

Answer: Fish obtain dissolved oxygen from water through their gills.

Q10. What is the difference between breathing and respiration?

Breathing	Respiration
It involves taking in and giving out air.	It releases energy from food inside cells.
It is a physical process.	It involves chemical reactions.
In humans, it mainly involves the lungs and air passages.	It occurs in living cells.

Think and Answer (Page 136)

Q. In what form is food stored in green plants?

Answer: Green plants commonly store extra food in the form of starch.

6. Growth

Q1. What is growth?

Answer: Growth is the increase in the size and mass of a living organism over time.

Q2. How do living things grow?

Answer: Living things grow through processes such as an increase in cell number and cell size.

Q3. How does a baby grow into an adult?

Answer: A baby grows as its cells increase in number and size, and its body develops over time.

Q4. How does a seed grow into a plant?

Answer: Under suitable conditions, a seed germinates and develops into a seedling. The seedling grows into a mature plant.

Q5. How is the growth of living things different from the growth of non-living things?

Answer: Living things grow through internal biological processes. Non-living things may become larger when material is added to their surfaces from outside.

Q6. Is the growth of a sugar crystal the same as the growth of a living organism?

Answer: No. A sugar crystal grows when more sugar particles are added to it from outside. A living organism grows through biological processes.

7. Excretion

Q1. What is excretion?

Answer: Excretion is the process by which living organisms remove metabolic waste products from their bodies.

Q2. Why is excretion necessary?

Answer: Excretion is necessary because some waste products can be harmful if they accumulate inside the body.

Q3. How do humans remove waste products?

Answer: Humans remove waste through urine, sweat, and exhaled air. Undigested food is removed through the anus by a separate process called egestion.

Q4. How do plants remove waste products?

Answer: Plants remove some waste products through stomata and other openings. Some wastes are stored in leaves, bark, or other parts that may later fall off.

Q5. What is the difference between excretion and egestion?

Answer: Excretion removes metabolic wastes produced by the body. Egestion removes undigested food from the digestive system.

Q6. Give an example of waste removal in plants.

Answer: Some plants release latex, while other wastes may be stored in leaves or bark.

8. Reproduction

Q1. What is reproduction?

Answer: Reproduction is the process by which living organisms produce new organisms of their own kind.

Q2. Why is reproduction important?

Answer: Reproduction is important for the continuation of a species from one generation to the next.

Q3. How do birds reproduce?

Answer: Birds reproduce by laying eggs, from which young birds develop.

Q4. How do cows reproduce?

Answer: Cows give birth to young ones.

Q5. How do plants reproduce?

Answer: Plants can reproduce through seeds, stems, roots, or other reproductive structures, depending on the plant species.

Q6. Give examples of different methods of reproduction in plants.

Answer:

- Pea and mango plants reproduce through seeds.
- Rose and some other plants can reproduce through stem cuttings.
- Potato can reproduce through buds or eyes on its tuber.
- Bryophyllum can reproduce through plantlets that develop along the edges of its leaves.

Q7. What is vegetative propagation?

Answer: Vegetative propagation is a type of asexual reproduction in plants in which new plants grow from vegetative parts such as stems, roots, or leaves.

Q8. What is the difference between reproduction in birds and cows?

Answer: Birds lay eggs, while cows give birth to young ones.

9. Movement and Response to Stimuli

Q1. What is movement?

Answer: Movement is a change in the position of an organism or one of its body parts.

Q2. What is a stimulus?

Answer: A stimulus is a change in the surroundings that causes an organism to respond.

Q3. What is a response?

Answer: A response is the reaction of an organism to a stimulus.

Q4. What is the plural of stimulus?

Answer: The plural of stimulus is stimuli.

Q5. How do animals respond to stimuli?

Answer: Animals respond to stimuli such as heat, light, sound, touch, and danger. For example, a person quickly withdraws their hand after touching a hot object.

Q6. How do plants respond to stimuli?

Answer: Plants respond to stimuli such as light, water, gravity, and touch. For example, a plant shoot bends towards sunlight.

Q7. What happens when a touch-me-not plant is touched?

Answer: Its leaves fold or close in response to touch.

Q8. Why does a plant near a window bend towards sunlight?

Answer: The shoot grows towards the light. This response is called phototropism.

Q9. How do roots respond to gravity?

Answer: Roots generally grow downwards in response to gravity. This is called positive gravitropism.

Q10. How do shoots respond to gravity?

Answer: Shoots generally grow upwards, away from the pull of gravity. This is called negative gravitropism.

Q11. How do roots respond to water?

Answer: Roots tend to grow towards areas where water is available. This response is called hydrotropism.

Q12. Do plants move even though they usually remain fixed in one place?

Answer: Yes. Plants show movements in their parts, such as shoots bending towards light and touch-me-not leaves closing when touched.

Try at Home: Plant Near a Window (Page 139)

Q. What happens when a potted plant is kept near a window where sunlight is available?

Answer: The plant shoot gradually bends towards the sunlight because it responds to light.

Try at Home: Touch-me-not Plant (Page 139)

Q. What happens when the leaves of a touch-me-not plant are touched?

Answer: The leaves fold or close in response to touch.

10. Identify the Process (Page 139)

Identify the life process described in each statement.

No.	Description	Answer
1	The increase in size and mass of an organism	Growth
2	The process of obtaining food by an organism	Nutrition
3	The process that helps release energy from food	Respiration
4	The process by which waste products are removed	Excretion
5	The process by which new organisms are produced	Reproduction

11. Quick Check: True or False (Page 139)

1. Plants do not show movement at all.

Answer: False. Plants show movements, such as shoots bending towards sunlight.

2. Reproduction is essential for the continuation of life on Earth.

Answer: True. Reproduction allows species to continue from one generation to the next.

3. A seed grows into a plant when it is sown in soil.

Answer: True, under suitable conditions. A seed needs appropriate conditions for germination and growth.

4. In plants, exchange of gases occurs through the gills.

Answer: False. Plants exchange gases mainly through stomata in their leaves and through other plant surfaces.

5. Removal of wastes from the body is not an essential process.

Answer: False. Waste removal is essential because the accumulation of metabolic wastes can harm an organism.

12. Differences Between Living and Non-living Things (Page 140)

The textbook compares living and non-living things using eight characteristics.

Characteristic	Living things	Non-living things
Cellular organisation	Cells are present.	Cells are absent.
Nutrition	Require food.	Do not require food.
Respiration	Undergo respiration.	Do not undergo respiration.
Growth	Show internal and irreversible biological growth.	Growth is absent; if present, it is external and reversible in the textbook comparison.
Excretion	Remove waste products.	Do not produce metabolic wastes.
Reproduction	Can reproduce as a species.	Do not reproduce.
Movement	Can show their own biological movements.	Do not move on their own biologically.
Response to stimuli	Respond to stimuli.	Do not show biological responses to stimuli.

Q1. What are the main differences between living and non-living things?

Answer: Living things have cells and perform life processes such as nutrition, respiration, growth, excretion, reproduction, movement, and response to stimuli. Non-living things do not perform all these life processes.

Q2. Do all living things move from one place to another?

Answer: No. Most animals can move from one place to another, but plants usually remain fixed in one place. Plants can still show movement in their parts.

Q3. Can all living organisms reproduce?

Answer: Reproduction is a characteristic of living organisms and is necessary for the continuation of a species. However, an individual organism may be unable to reproduce, for example because of age or infertility.

Q4. What similarities do living and non-living things share?

Answer: Both occupy space and have mass. Both are made of basic structural units: living things are made of cells, while non-living matter is made of atoms and molecules.

13. Essential Conditions for Germination of a Seed

Q1. What is a seed?

Answer: A seed is a plant structure that contains an embryo and can develop into a new plant under suitable conditions.

Q2. What is germination?

Answer: Germination is the process in which a seed begins to grow and develops into a seedling under suitable conditions.

Q3. What is an embryo in a seed?

Answer: An embryo is the young, undeveloped plant present inside a seed.

Q4. What conditions are needed for seed germination?

Answer: Seeds generally need water, air, and a suitable temperature for germination. The role of sunlight depends on the plant species and the stage of growth.

A. Water

Q5. Why is water necessary for germination?

Answer: Water softens the seed coat and activates the processes needed for the embryo to grow. It also helps make stored food available to the developing embryo.

B. Air

Q6. Why is air necessary for seed germination?

Answer: Seeds need oxygen from the air for respiration, which releases energy for growth.

C. Suitable Temperature

Q7. Why is a suitable temperature necessary for germination?

Answer: A suitable temperature allows the enzymes and other processes needed for germination to work properly. Very low or very high temperatures may prevent or slow germination.

Q8. Do all seeds need sunlight to germinate?

Answer: No. Many seeds can germinate without sunlight if they have enough water, oxygen, and a suitable temperature. Sunlight is usually important for healthy growth after the seedling emerges, although requirements differ among plant species.

Q9. Why does a dry seed not germinate immediately?

Answer: A dry seed generally remains inactive until it receives sufficient water and the other necessary conditions are available.

Q10. What is the function of stored food in a seed?

Answer: Stored food supplies energy and nutrients to the embryo during germination and early growth.

Think and Answer (Page 141)

Q. Is a seed living or non-living? Explain.

Answer: A viable seed is living because it contains a living embryo. It may remain dormant until suitable conditions allow it to germinate.

14. Activity: Conditions Required for Germination (Page 142)

The textbook activity compares four pots under different conditions.

- Pot I: Seed receives sunlight, air, and water.

- Pot II: Seed receives air and water but is kept away from direct sunlight.
- Pot III: Seed receives sunlight and air but no water.
- Pot IV: Seed receives sunlight and water but has very little or no suitable air supply because the soil is kept excessively wet.

Q1. Which pot is expected to show germination under the stated conditions?

Answer: Pots I and II are expected to show germination if the temperature is suitable and the seeds are viable. Sunlight is not essential for the initial germination of many common seeds.

Q2. Why will the seed in Pot III fail to germinate normally?

Answer: It does not receive water, which is essential for germination.

Q3. Why may the seed in Pot IV fail to germinate normally?

Answer: Excess water can reduce the air available to the seed. Without enough oxygen, respiration and germination are affected.

Q4. What is the conclusion of this activity?

Answer: Water, air, and a suitable temperature are necessary for normal germination. Sunlight is not essential for the initial germination of many seeds.

15. Growth and Movement in Plants

Q1. How does a plant shoot respond to sunlight?

Answer: The shoot grows towards sunlight.

Q2. How do roots grow when a plant is kept upright?

Answer: Roots generally grow downwards in response to gravity.

Q3. What happens if a germinating seed is placed upside down?

Answer: The root bends and grows downwards, while the shoot bends and grows upwards, regardless of the seed's initial orientation.

Q4. What happens when light falls on a plant from one side?

Answer: The shoot generally bends towards the light, while the root tends to continue growing downwards.

Q5. What is phototropism?

Answer: Phototropism is the growth response of a plant part towards or away from light.

Q6. What is gravitropism?

Answer: Gravitropism is the growth response of a plant part to gravity.

Q7. What is hydrotropism?

Answer: Hydrotropism is the growth response of a plant part to water. Roots commonly grow towards moisture.

Think and Answer (Page 143)

Q. If a plant is put upside down, what would be the direction of growth of its roots and shoots?

Answer: The roots will bend and grow downwards, towards gravity. The shoots will bend and grow upwards, away from gravity.

Activity: Direction of Growth (Page 144)

The textbook compares seedlings exposed to light from different directions and seedlings kept inside a box with a side opening.

Q1. What is the aim of this activity?

Answer: The aim is to observe how light and gravity influence the growth of plant shoots and roots.

Q2. What is expected when a seedling receives light from one direction?

Answer: Its shoot generally bends towards the light.

Q3. What is expected when a seedling receives light from all directions?

Answer: Its shoot generally grows upwards, provided other conditions are suitable.

Q4. What is expected when a seedling receives no light?

Answer: The seedling may still begin growing by using stored food, but it usually becomes pale and weak if it remains without light.

Q5. What did Jagadish Chandra Bose contribute to the study of plants?

Answer: Jagadish Chandra Bose studied how plants respond to stimuli. He developed instruments, including the crescograph, to measure plant growth and demonstrated that plants respond to environmental changes.

16. Life Cycle

Q1. What is a life cycle?

Answer: A life cycle is the series of stages through which a living organism passes during its life, from the beginning of life to adulthood and eventually death.

Q2. Why is a life cycle important?

Answer: A life cycle describes how an organism develops, grows, and reproduces, allowing its species to continue.

Q3. Do plants and animals have life cycles?

Answer: Yes. Both plants and animals pass through different stages of development during their lives.

17. Life Cycle of a Plant

Q1. What are the main stages in the life cycle of a flowering plant?

Answer: The main stages are seed, germination, seedling, mature plant, flowering, fruit formation, seed formation, and the beginning of a new generation.

Q2. What happens during germination?

Answer: The embryo inside the seed begins to grow, and the seed develops into a seedling.

Q3. What happens when a plant becomes mature?

Answer: A mature plant may produce flowers, which can develop into fruits containing seeds after successful reproduction.

Q4. How does a plant life cycle continue?

Answer: Seeds produced by a mature plant can germinate and grow into new plants under suitable conditions.

Q5. What happens when a plant does not get the conditions needed for growth?

Answer: Its growth may slow down or stop. If the conditions remain unsuitable, the plant may eventually die.

18. Life Cycle of a Mosquito

Q1. What are the stages in the life cycle of a mosquito?

Answer: Egg → Larva → Pupa → Adult mosquito.

Q2. Where do mosquitoes commonly lay their eggs?

Answer: Many mosquitoes lay their eggs in or near water. Different mosquito species have different egg-laying habits.

Q3. What is a mosquito larva?

Answer: A larva is the immature stage that hatches from a mosquito egg and lives in water in many mosquito species.

Q4. What is a mosquito pupa?

Answer: A pupa is the stage between the larva and the adult mosquito. The adult mosquito develops inside the pupal stage.

Q5. What happens after the pupal stage?

Answer: An adult mosquito emerges from the pupa.

Q6. Why are mosquitoes considered harmful?

Answer: Some mosquitoes spread diseases such as malaria, dengue, and chikungunya.

Q7. How can we reduce mosquito breeding?

Answer: We should prevent water from collecting in containers, tyres, pots, coolers, and other places. Water storage containers should be covered, and stagnant water should be removed or properly managed.

Think and Answer (Page 145)

Q. Why is it advised to get rid of stagnant water?

Answer: Stagnant water provides breeding places for mosquitoes. Removing it helps reduce mosquito breeding and the risk of mosquito-borne diseases.

Think and Answer (Page 146)

Q. You may have seen mosquito larvae and pupae frequently on the surface of water bodies. What would be the reason?

Answer: Many mosquito larvae and pupae live in water and come to the surface to breathe air. This is why they are often seen near the water's surface.

19. Life Cycle of a Frog

Q1. What are the main stages in the life cycle of a frog?

Answer: Egg → Embryo → Tadpole → Froglet → Adult frog.

Q2. What is spawn?

Answer: Spawn is a cluster or mass of frog eggs laid in water.

Q3. What is a tadpole?

Answer: A tadpole is the aquatic larval stage of a frog. It has a tail and initially breathes mainly through gills.

Q4. What is a froglet?

Answer: A froglet is a young frog that has developed legs and is gradually changing into an adult frog. Its tail becomes shorter as development continues.

Q5. How does a tadpole change into an adult frog?

Answer: The tadpole develops hind legs and then forelegs. Its lungs develop, and its tail gradually disappears. It becomes a young frog and then an adult frog.

Q6. What is metamorphosis?

Answer: Metamorphosis is the process in which an animal undergoes major changes in body structure during its development.

Q7. How is the life cycle of a frog different from that of a mosquito?

Answer: A frog develops through the egg, tadpole, froglet, and adult stages. A mosquito develops through the egg, larva, pupa, and adult stages.

Q8. Why do frogs lay their eggs in water?

Answer: Frog eggs generally need a moist or aquatic environment for development. Their eggs do not have the hard protective shells found in birds' eggs.

20. Important Comparison Questions

A. Life Cycle of a Plant and a Frog

Plant	Frog
Begins with a seed in flowering plants.	Begins with an egg.
A seed germinates into a seedling.	An egg develops into an embryo and then a tadpole.
The plant grows and may produce flowers and fruits.	The tadpole develops into a froglet and then an adult frog.
A mature plant may produce new seeds.	An adult frog reproduces and lays eggs.

B. Life Cycle of a Mosquito and a Frog

Mosquito	Frog
Egg → Larva → Pupa → Adult	Egg → Embryo → Tadpole → Froglet → Adult
Has a pupal stage.	Does not have a pupal stage.
Larva and pupa are immature stages.	Tadpole is the larval stage.
Many species breed in or near water.	Frogs generally lay eggs in water.

C. Characteristics of Living Things

Characteristic	Example
Nutrition	A plant makes food by photosynthesis.
Respiration	A human releases energy from food.
Growth	A seedling grows into a mature plant.
Excretion	Kidneys remove waste products from blood.
Reproduction	A hen lays eggs.
Movement	A person walks; a shoot bends towards light.
Response to stimuli	Touch-me-not leaves close when touched.
Cellular organisation	A human body is made up of cells.

21. Additional Important Questions for Examination

Q1. Why is a plant considered a living organism even though it does not walk?

Answer: A plant is living because it grows, respires, needs nutrition, removes waste, reproduces, and responds to stimuli.

Q2. Why is a seed considered dormant rather than dead?

Answer: A viable dormant seed contains a living embryo that can begin growing when suitable conditions become available.

Q3. What is the difference between a stimulus and a response?

Answer: A stimulus is a change that causes a reaction. A response is the reaction to that change.

Q4. Why do living organisms need energy?

Answer: They need energy for growth, movement, repair, reproduction, and other life processes.

Q5. Why is respiration necessary even in plants that make their own food?

Answer: Photosynthesis stores energy in food, while respiration releases usable energy from food for the plant's life processes.

Q6. Why can a plant not grow properly without water?

Answer: Water is needed for germination, transport of substances, photosynthesis, and maintaining the firmness of plant cells.

Q7. Why is air important for germination?

Answer: Oxygen in air is required for respiration, which releases the energy needed by the growing embryo.

Q8. How does a plant respond to touch?

Answer: Some plants, such as touch-me-not, fold their leaves when touched.

Q9. What is the role of an embryo in a seed?

Answer: The embryo develops into a new plant during germination.

Q10. Why is the mosquito life cycle important to understand?

Answer: Understanding the mosquito life cycle helps us identify and remove breeding sites and reduce the spread of mosquito-borne diseases.

Q11. What is the role of the larval stage in a frog's life cycle?

Answer: The tadpole grows and develops the body structures needed for life as an adult frog.

Q12. Why is reproduction important for a species but not necessary for every individual to survive?

Answer: Reproduction ensures the continuation of a species over generations. An individual organism can live without reproducing, but a species cannot continue indefinitely without reproduction.

22. Multiple-Choice Questions (MCQs)

Chapter Practice Quiz

0/15 answered

1. What is the basic structural and functional unit of life?

A. Organ
B. Tissue
C. Cell
D. Organ system

2. Which of these is a unicellular organism?

A. Dog
B. Mango tree
C. Amoeba
D. Human

3. Which process releases energy from food?

A. Growth
B. Respiration
C. Reproduction
D. Excretion

4. Which gas is used in aerobic respiration?

A. Oxygen
B. Nitrogen
C. Hydrogen
D. Helium

5. Which process removes metabolic waste products?

A. Nutrition
B. Reproduction
C. Excretion
D. Growth

6. Which plant reproduces through plantlets on its leaves?

A. Mango
B. Bryophyllum

C. Pea
D. Wheat

7. What is a stimulus?

A. A type of food
B. A change that causes a response
C. A cell
D. A waste product

8. Which plant commonly responds to touch by folding its leaves?

A. Mango
B. Rose
C. Touch-me-not
D. Pea

9. Which three conditions are generally needed for seed germination?

A. Water, air, suitable temperature
B. Sunlight, soil, fertiliser only
C. Water, darkness, cold
D. Soil, stones, sunlight

10. What is the first stage in the mosquito life cycle?

A. Pupa
B. Adult
C. Larva
D. Egg

11. Which is the larval stage of a frog?

A. Pupa
B. Tadpole
C. Froglet
D. Adult

12. Which stage comes after the larva in a mosquito's life cycle?

A. Egg
B. Adult
C. Pupa
D. Tadpole

13. What is the process by which a seed starts growing?

A. Pollination
B. Germination
C. Excretion
D. Condensation

14. Why should stagnant water be removed?

A. To help mosquitoes breed
B. To reduce mosquito breeding
C. To increase humidity
D. To make water salty

15. What is the usual direction of root growth in response to gravity?

A. Upwards
B. Sideways only
C. Downwards
D. Towards darkness only
Check Answers

23. Fill in the Blanks

1. The basic structural and functional unit of life is the cell.
2. A group of similar cells forms a tissue.
3. The process of obtaining food is called nutrition.
4. The release of energy from food is called respiration.
5. The removal of metabolic waste products is called excretion.
6. The process of producing new organisms is called reproduction.
7. A change that causes an organism to react is called a stimulus.
8. The growth of a plant part towards light is called phototropism.
9. The growth response to gravity is called gravitropism.
10. The process in which a seed begins to grow is called germination.
11. A young mosquito that lives in water is called a larva.
12. The stage between the mosquito larva and adult is called the pupa.
13. The larval stage of a frog is called a tadpole.
14. The series of stages in an organism's life is called its life cycle.
15. Plants exchange gases through small openings called stomata.

24. True or False

1. All living organisms are made up of cells. — True
2. Yeast is a multicellular organism. — False
3. Plants perform respiration. — True
4. Non-living things perform all the life processes. — False
5. Plants can respond to light and touch. — True

6. Every seed needs direct sunlight to begin germination. — False
7. Water and oxygen are generally needed for seed germination. — True
8. A mosquito has an egg, larva, pupa, and adult stage. — True
9. A frog's larval stage is called a tadpole. — True
10. Removing stagnant water can help reduce mosquito breeding. — True
11. Excretion and egestion mean exactly the same thing. — False
12. The roots of most plants grow downwards in response to gravity. — True
13. Reproduction helps continue a species. — True
14. A dormant, viable seed is necessarily dead. — False
15. A plant shoot usually grows towards light. — True

25. Quick Revision Notes

- Cell: The basic structural and functional unit of life.
- Tissue: A group of similar cells performing a specific function.
- Organ: A structure made of tissues working together.
- Nutrition: Obtaining or making food.
- Respiration: Releasing energy from food.
- Growth: An increase in size and mass through biological processes.
- Excretion: Removal of metabolic waste.
- Reproduction: Production of new organisms.
- Stimulus: A change that causes a response.
- Germination: The beginning of growth of a seed.
- Conditions for germination: Water, air, and suitable temperature.
- Phototropism: Growth response to light.
- Gravitropism: Growth response to gravity.
- Hydrotropism: Growth response to water.
- Plant life cycle: Seed → Germination → Seedling → Mature plant → Flower/fruit/seed.
- Mosquito life cycle: Egg → Larva → Pupa → Adult.
- Frog life cycle: Egg → Embryo → Tadpole → Froglet → Adult.

These notes cover the questions visible in the uploaded textbook pages 134–146, including the diagrams of cellular organisation, seed germination, plant growth, and the mosquito and frog life cycles. The uploaded file does not include a separate end-of-chapter exercise page.

10-living-creatures-exploring-their-characteristics.pdf